

Evaluating an LLM-Powered Chatbot for Cognitive Restructuring: Insights from Users and Mental Health Professionals

YINZHOU WANG, William & Mary, USA

YIMENG WANG, William & Mary, USA

YE XIAO, William & Mary, USA

LIABETTE ESCAMILLA, William & Mary, USA

BIANCA AUGUSTINE, William & Mary, USA

KELLY CRACE, University of Virginia, USA

GANG ZHOU, William & Mary, USA

YIXUAN ZHANG, William & Mary, USA

Recent advances in large language models (LLMs) create opportunities to expand access to mental healthcare. An emerging body of work focuses on LLM-based cognitive restructuring (CR), yet existing systems are often limited to generating alternative beliefs or engaging in short, scripted exchanges, offering little of the sustained interactivity required for effective restructuring. In this work, we present CRBot, an LLM-powered chatbot designed with mental health professionals to deliver CR. We conducted a two-week user study with 19 participants, complemented by expert review of conversation transcripts. Results show that CRBot can scaffold the core steps of CR, support self-reflection through Socratic questioning, and provide effective psychoeducation. At the same time, challenges arose, including rigidity, oversimplified questioning, pathologization of appropriate emotions, and misinterpretation of nuanced context. We conclude with design implications for building more interactive, flexible, and ethically robust LLM-enabled CR systems.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**.

Additional Key Words and Phrases: Large Language Models, Human-AI-Interaction, Mental health, Cognitive Restructuring

1 Introduction

Recent advances in large language models (LLMs) have the potential to expand access to mental health interventions further, offering support that is available anytime and anywhere [20]. Such potential is particularly relevant given the global shortage of mental health professionals, limited patient-therapist time, and barriers to care such as cost [50]. LLMs have demonstrated certain therapist-like abilities, such as delivering psychoeducation [33] and adhering to intervention protocols [41], laying the groundwork for personalized support through user-friendly chatbots. Given the potential of LLMs, researchers have begun to explore the design of LLM-powered tools in self-help mental health applications. A lot of these applications are often “grounded” or inspired by existing

Authors' Contact Information: Yinzhou Wang, William & Mary, Williamsburg, Virginia, USA, ywang143@wm.edu; Yimeng Wang, William & Mary, Williamsburg, Virginia, USA, ywang139@wm.edu; Ye Xiao, William & Mary, Williamsburg, Virginia, USA, yxiao03@wm.edu; Liabette Escamilla, William & Mary, Williamsburg, Virginia, USA, laescamilla@wm.edu; Bianca Augustine, William & Mary, Williamsburg, Virginia, USA, braugustine@wm.edu; Kelly Crace, University of Virginia, Charlottesville, Virginia, USA, kelly.crace@virginia.edu; Gang Zhou, William & Mary, Williamsburg, Virginia, USA, gzhou@wm.edu; Yixuan Zhang, William & Mary, Williamsburg, Virginia, USA, yzhang104@wm.edu.

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therapeutic approaches and theories, such as cognitive behavioral therapy (CBT) [8]. CBT is a widely validated approach for addressing diverse mental health conditions by restructuring maladaptive cognitions and behaviors through systematic, evidence-based strategies. One of its core techniques is *cognitive restructuring* (CR) [8], which targets identifying and challenging distorted thoughts to promote more accurate or beneficial perspectives.

The highly structured nature of the CR approach makes it easier to implement and ensures consistency in application across diverse contexts. Accordingly, an emergent body of studies has begun exploring how to design LLM-powered CR systems [16, 22, 28, 36, 37, 51, 53]. The goals of CR include questioning long-held maladaptive beliefs and considering the precision and utility of alternative perspectives [8]. To achieve these goals, the implementation of CR typically relies on, but is not limited to, using Socratic questioning to help the client determine the validity of the belief, finding possible alternative interpretations, and decatastrophizing the situation—processes that demand high interactivity [2]. For LLM-based approaches, this requires multiple conversational turns and open-ended questioning. However, existing LLM-based CR systems focus either on the generation of alternative beliefs [36, 37, 51] or the development of “brief-turns” conversational CR (i.e., conversations with a few messages exchanged) [28, 53]. Moreover, due to the uncontrollable nature of LLMs [13, 45], poor representation of psychotherapy transcripts in its training data, and ethical concerns regarding using therapeutic transcripts for training [40], alternative human–LLM interaction approaches are needed. Therefore, we ask: **how should interactive LLM-based CR systems be designed?**

In this work, we iteratively designed an LLM-powered chatbot, CRBot, in collaboration with mental health professionals (MHPs). We introduced the idea of “adaptive scaffolding”, *a structured, collaborative process where the conversational flow and agent behavior adjust to users’ progress in exploring their thoughts*. In the context of human–LLM interaction for CR, this approach maintains adherence to the CR structure while fostering active user learning. We then conducted a user study with 19 participants. Findings showed that users valued the scaffolding approach for providing step-by-step guidance and fostering self-awareness, but also noted limitations around flexibility, pacing, and oversimplified questioning. Most participants appreciated CRBot’s compassionate stance, though some highlighted risks of invalidation, and while psychoeducational features were praised, users suggested refinements to improve applicability. To complement these insights, four MHPs reviewed the conversation logs to identify CRBot’s alignments and misalignments with real-world practice. Their analysis underscored CRBot’s ability to follow core CBT principles, maintain a natural conversational flow, and pose Socratic questions, while also revealing risks: misuse of positive regard, power imbalances introduced through leading or evaluative language, and contextual understanding gaps that occasionally led to misinterpreted user states.

Collectively, these findings illustrate both the promise and pitfalls of applying LLMs to cognitive restructuring. On one hand, our study highlights how “adaptive scaffolding” can enforce CR protocol and support reflective learning; on the other hand, it surfaces areas where CRBot diverges from real-world therapeutic practice. We conclude by discussing implications for the design of future LLM-based CR systems across four dimensions: balancing structured guidance, psychoeducation, and flexibility in adaptive scaffolding; refining the expression and recognition of emotion, including positive regard and contextual sensitivity; navigating power dynamics and preserving user autonomy; and moving toward safer LLM-based CR through rethinking evaluation strategies and strengthening human oversight.

In this work, we contribute: **1)** an evaluation of an LLM-powered chatbot for cognitive restructuring with 19 participants and four mental health professionals, and **2)** insights and design implications that can guide future research and development in LLM-based cognitive restructuring applications.

2 Background & Related Work

We first provide some domain background focusing on psychotherapy, cognitive behavioral therapy (CBT), and cognitive restructuring (CR) to provide contextual information that helps situate our work. Then, we describe related work, discussing the status of existing LLM-powered CR systems.

2.1 Background on Psychotherapy, Cognitive Behavioral Therapy, and Cognitive Restructuring

Psychotherapy contains a variety of interventions designed to alleviate psychological distress and improve mental well-being [29]. Treatment approaches vary widely, but they often emphasize building a trusted therapeutic relationship in which clients can safely explore their thoughts, emotions, and behaviors. Among these approaches, **Cognitive behavioral therapy (CBT)** is one of the most widely studied and empirically supported treatments for various mental health conditions, such as depression [15] and anxiety [47]. A hallmark of CBT is its structured, goal-oriented methodology, focusing on the interplay between thoughts, emotions, and behaviors. By identifying and modifying maladaptive cognitions, CBT interventions aim to produce meaningful shifts in emotional regulation and behavioral patterns [1]. One of CBT's core techniques is **cognitive restructuring (CR)**, which guides clients to identify, challenge, and replace maladaptive thought patterns with more positive or beneficial alternatives [8]. Typically, CR proceeds through three interconnected steps: (1) *Identify distorted thoughts*: The client is guided to recognize triggering situations and maladaptive thoughts (e.g., "Can you share what's been on your mind lately?"), (2) *Challenge distorted thoughts*: The therapist and client collaboratively question the validity of these thoughts (e.g., "Do you have any concrete evidence to support the thought that others would think you're weird?"), and (3) *Develop alternative thoughts*: The client is encouraged to replace maladaptive thoughts with more rational or fact-based alternatives (e.g., "Can you try to reframe this thought in a way that's based on facts rather than assumptions?").

2.2 The Use of LLMs in Cognitive Restructuring

Recent research on CR [16, 22, 28, 36, 37, 51, 53] using LLMs spans various approaches, such as performance comparisons between humans and AI, in-context learning for generating reframed thoughts, and developing "brief-turns" conversational CR. For example, Li et al. [22] compared human and GPT-4 performance on reframing negative vignettes, while Sharma et al. [36, 37] utilized retrieval-enhanced in-context learning to generate reframed thoughts from identified negative thoughts and leading situations. The authors also introduced a framework of seven linguistic attributes for thought restructuring. Wang et al. [51] further explored this line of work by leveraging GPT-3 to refine reframed thoughts iteratively.

Some other researchers implemented conversation-based CR [16, 28, 53]. While these systems implemented all CR stages, the conversation usually takes only a few turns and lacks interactivity. In particular, Nie et al. designed a multi-agent system to deliver a questionnaire-style CR with limited conversational flexibility. At each stage, the chatbot delivers predefined questions, evaluates users' responses, and provides guidance if responses are determined invalid. Based on the guidance, the users adjust their previous response and repeat the process [28]. Xiao et al. trained an LLM on a synthetic CR dataset, and their case studies indicate that the LLM delivers three-round conversations, with each round corresponding to a stage of CR [53].

Together, this body of research validated the feasibility of LLM-enabled CR and contributed to the nuanced understanding of thought reframing in human-LLM-interaction. Nevertheless, the goal of cognitive restructuring is to encourage clients to question deeply ingrained, unhelpful beliefs (e.g., "People will notice I'm anxious and think there is something wrong with me") and to consider the accuracy and utility of an alternative perspective (e.g., "People might notice I'm a little anxious but consider it unimportant") [8]. While current studies have focused on the accuracy and utility of alternative perspectives, none have implemented sustained, interactive processes,

such as multi-turn Socratic questioning, iterative perspective-taking, and decatastrophizing, that are critical for guiding clients through deeper exploration of their thoughts (see an expert-crafted example in Appendix A.2).

Addressing this gap requires approaches that align LLM behavior with therapeutic principles while supporting sustained interactivity. Although fine-tuning LLMs on therapeutic dialogues would be ideal, such data are scarce in general-purpose training corpora and are further constrained by privacy and ethical concerns [40]. Moreover, existing CBT datasets are either unrelated to CR or were primarily designed for classification tasks. For instance, Bhatt et al. [4] listed 21 datasets, most of which involve modalities such as audio, facial, and physiological signals intended for cognitive behavioral analysis. Some textual datasets capture thinking errors, emotions, situations, and reframes, which are related to CR [34, 57]. However, these resources are not conversational in nature and therefore cannot be used to fine-tune an LLM for CR dialogues. Given the scarcity of realistic CR training data and associated ethical concerns, prompt engineering represents a practical starting point for aligning LLM behavior with therapeutic principles. Therefore, we started with a zero-shot prompt and iteratively designed and evaluated an adaptive scaffolding approach for CR, highlighting interactivity, adherence to protocol, and reflective learning.

3 Methods

In this section, we describe our study procedure, including an iterative tool design process with five MHPs to create the chatbot CRBot (see Section 3.1) and a user study with 19 participants to interact with the chatbot in real-world scenarios (see Section 3.2), and an interview study with four MHPs to review the conversation transcripts, providing professional perspectives and identifying potential risks (see Section 3.3). This research was approved by our institution's Institutional Review Board (IRB).

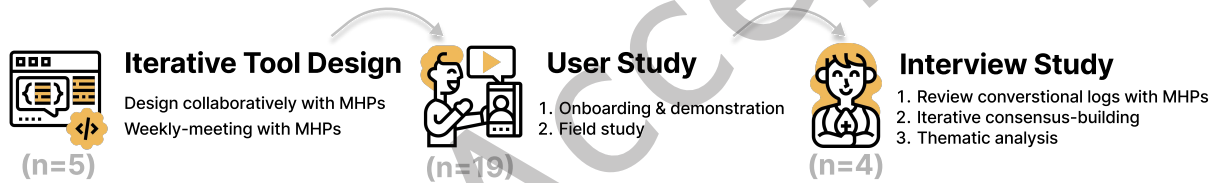


Fig. 1. Overall study flow

3.1 Iterative Tool Design

3.1.1 Co-design Process. While existing conversational CR systems frequently lack interactivity, we identified one study that used zero-shot prompting to implement longer CR conversations [16]. Although the overall results of their intervention were reported as positive, the paper did not analyze the specific strengths and limitations regarding cognitive restructuring. To explore potential opportunities, we adapted their prompt (see Appendix A.3.1), implemented an initial web-based chatbot prototype, and conducted an initial evaluation with MHPs.

We first invited five MHPs to evaluate the zero-shot implementation in the first interview (60 minutes). The MHPs had an average of 15 years of clinical counseling experience, ranging from 5 to 35 years. All remained actively involved in day-to-day mental health counseling, maintaining full caseloads. Demographically, the group comprised one man and four women, with two identifying as African American or Black and three identifying as White. The lead author provided a brief demonstration of the prototype, described its core tasks, and walked through the user interface. Following the demonstration, MHPs independently explored the prototype, role-played as a client, while sharing their screens. The think-aloud protocol [49] was used to allow for articulating their immediate impressions, concerns, and suggestions.

Our first evaluation resulted in three important observations (see examples in Appendix A.1):

- (1) While all MHPs agreed that the zero-shot prototype covered all stages of CR, three MHPs noted that **the stages were explored only superficially**. For example, one MHP commented, “*I think this [Stage 3: Develop alternative thoughts] came up too quickly. [The LLM] needs more work that gets them into challenging that thought.*” This observation underscored a limitation of using a single zero-shot prompt to span the entire CR process, as it risks premature transitions between stages. Accordingly, we derived our first design principle: **ensuring sufficient exploration within each CR stage**. To address this, we introduced stage-specific prompts that maintain alignment with the CR procedure and minimize shallow or rushed progression.
- (2) One MHP intentionally role-played as a healthy individual expressing rational thoughts. In this scenario, the **system rigidly insisted on locating a distortion, even when none was present**. Such behavior not only undermines the usability of the system but also raises potential risks of pathologizing normal thoughts. Thus, this observation led to our second design principle: **handle edge case scenarios to avoid inappropriate probing**. We then utilized few-shot prompting, assigning an example for each edge case.
- (3) Three MHPs indicated that more psychoeducation is needed, including the type and explanation of distorted thoughts. For example, one MHP said “*Need more explanations for the concepts [cognitive distortions] to clarify for the clients.*” Another MHP described how they guide clients through the CR practice “*If they are like, seem to get stuck and they’re like, ‘No, I’m really don’t know what’s my distorted thoughts are’, [I’d say], Okay, well, let’s just take a look at our sheet of distorted thoughts, and what the definitions are, and take a look again and see if we think that’s a factual thought or a distorted thought.*” This reflection highlights that psychoeducation is not only about labeling distortions but also about equipping clients with references they can use to work through distortions themselves, while the therapist provides guidance rather than direct answers. This led to the last design principle: **integrate psychoeducation with user autonomy**. To operationalize this, we included scaffolding questions alongside distortion types and definitions, prompting users toward active learning.

We then used these principles to guide the iterative co-design with MHPs. Weekly online sessions followed a consistent procedure: demonstrating new features, inviting MHPs to interact individually with the updated prototype, and collecting structured feedback on usability, strengths, and limitations. During each session, MHPs carefully evaluated whether the prototype’s outputs aligned with the agreed design principles. Whenever misalignments arose, they provided targeted feedback, revised prompts, or suggested additional examples to correct the behavior. This cycle of testing and adjustment continued until no further misalignments were observed, contributing to a balance of interactivity, protocol adherence, and psychoeducation, leading to the finalized design of CRBot.

3.1.2 CRBot and Implementations. CR is delivered with adaptive scaffolding by two agents, a *Dialogue Facilitator* and a *Scaffolding Coordinator*. Figure 2 shows an example. We include all prompts and demonstrations of a few-shot examples in Appendix A.3.

The Dialogue Facilitator Agent is responsible for facilitating conversation, including asking questions, providing empathetic responses, and providing normalization statements throughout all stages. The speaker agent’s prompts include system prompts: “You are a cognitive behavior therapist, and your job is to ...” and a few examples. We asked our MHPs to create these examples, and to ensure the examples cover edge cases, we also asked them to vary their examples based on different scenarios:

- (1) Successful completion: The user completes each stage without any struggle. Thus, they successfully identify the negative thought, challenge it, and generate a rational one.

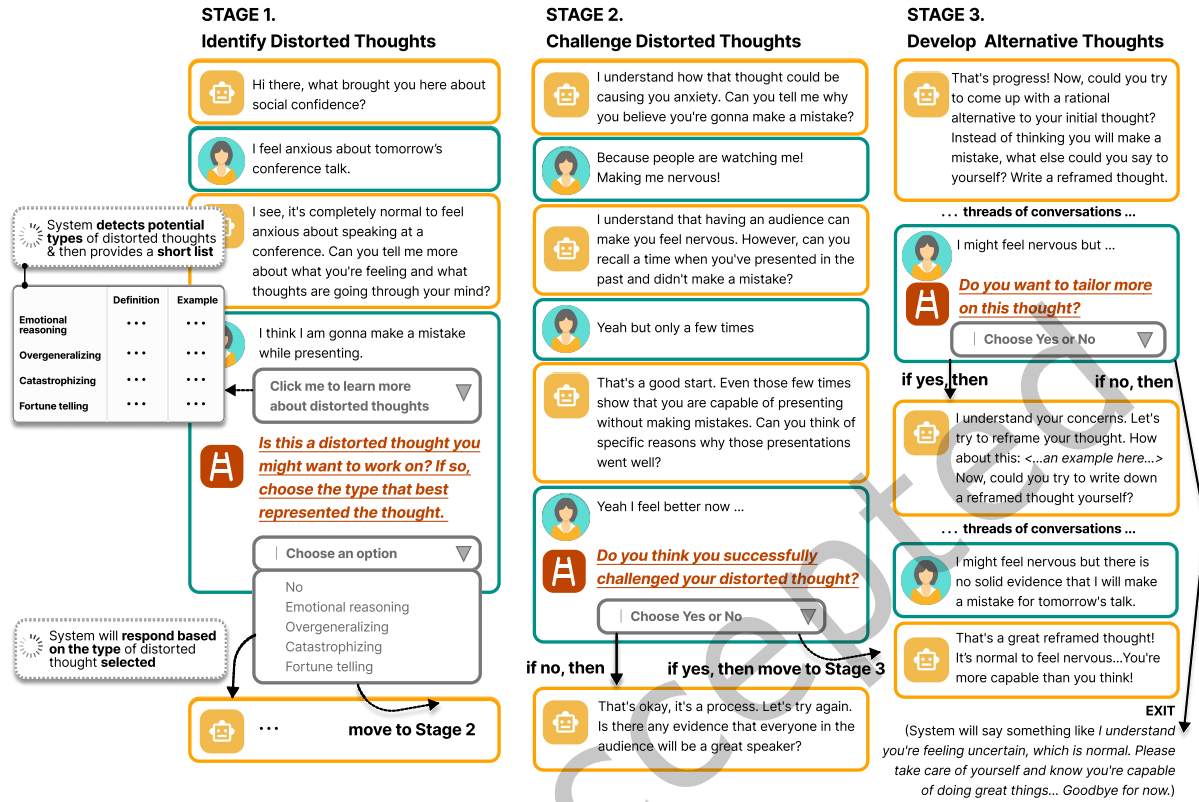


Fig. 2. A simplified illustration of the logic and conversational flow within the CR exercise based on the real outputs from CRBot. CRBot prompts the user to **identify distorted thoughts** (STAGE 1), and **challenge distorted thoughts** (STAGE 2) through guided interactions. The process involves the initial identification of context, selection of distorted thought types, provision of psychoeducation, and guided attempts to restructure distorted thoughts and **develop alternative thoughts** (STAGE 3) into more rational and balanced perspectives. CRBot also fosters self-reflection and the development of cognitive restructuring skills. Underlined italic text in dark orange represents examples of our scaffolding questions.

- (2) Absence of negative thoughts: The user does not have any negative thoughts. For example, the user has a rational understanding of their situation.
- (3) Identification challenges: The user fails to identify the negative thoughts. For example, the user does not think their thought is distorted (but it is from the therapist's perspective).
- (4) Challenging barriers: The user can't challenge their negative thoughts. For example, the user can't provide evidence against the negative thoughts.
- (5) Creation of alternative thoughts: The user fails to develop a rational alternative thought.

Adaptive Scaffolding Approach is accomplished through the collaboration between the user, the dialogue facilitator, and the scaffolding coordinator (Figure 3). The dialogue facilitator uses normalization, validation, and Socratic questioning to maintain engagement and enable in-depth thought exploration. When specific information, e.g., a distorted thought, is detected, the scaffolding coordinator poses targeted scaffolding questions. After the user reflects on and responds to the question, the scaffolding coordinator communicates the user's

decision (e.g., distortion type - “Fortune telling”) to the dialogue facilitator. The prompts for both agents are then adjusted to align with the user’s choice.

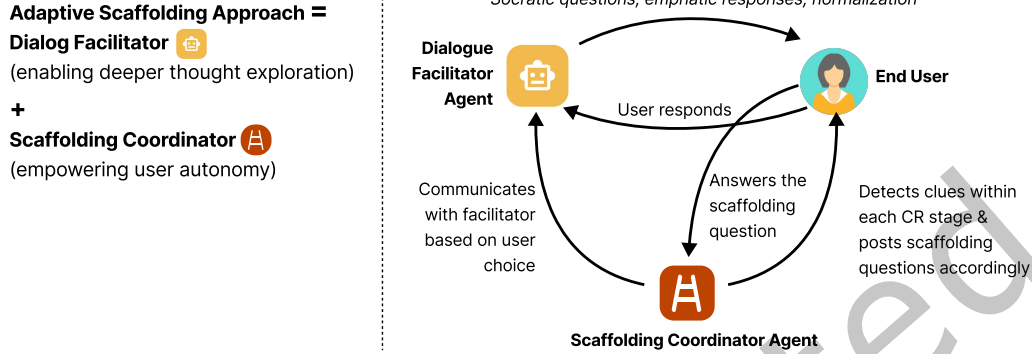


Fig. 3. An illustration of the Adaptive Scaffolding Approach, which integrates a dialogue facilitator that enables deeper thought exploration and a scaffolding coordinator Agent that empowers user autonomy by dynamically adjusting to user responses. The facilitator and coordinator work together to guide the end user through cognitive restructuring stages with socratic questions, empathetic responses, and normalization techniques, which seek to enhance the therapeutic interaction.

The Scaffolding Coordinator Agent is integrated into the conversational flow to prompt users to reflect on their responses and assist them in deciding whether to move on to the next stage. CRBot uses three types of scaffolding questions across different CR stages:

- (1) During Stage 1 (identifying distorted thoughts), the agent asked: “*Is this a distorted thought you might want to work on? If so, choose the type that best represents the thought.*”
- (2) During Stage 2 (challenging distorted thoughts), the agent asked: “*Do you think you successfully challenged your negative thoughts?*”
- (3) During Stage 3 (developing alternative thoughts), the agent asked: “*Do you want to tailor more on this thought?*”

The Scaffolding Coordinator agent’s role is to maintain structured progression through the CR stages by dynamically adjusting to user responses. By prompting reflection and checking alignment with the current stage, the Coordinator helps prevent conversational drift and guides transitions to occur when the user has provided sufficient evidence to progress. It monitors for the presence of a distorted thought, checks whether the user has offered enough evidence to challenge the distortion, and assesses whether a valid alternative thought has been developed. We used GPT-4 and designed three prompts containing a few examples to accomplish these tasks.

3.2 User Study

Participants Recruitment & Overview. We leveraged a participant pool from the research team’s previous projects on mental health. To ensure participant safety and well-being while gathering meaningful user feedback, we reference self-report widely-used mental health questionnaires (e.g., PHQ-9 [19] and GAD-7 [39] scores). Participants with scores indicating “severe” levels of distress are deemed unsuitable for the experimental research. We sent email and text invitations, including a screener survey, descriptions of the research purpose, study procedures, and a consent form, to potential participants ($n = 150$). The screener survey included self-report

mental health questionnaires and demographics. Detailed questions of the screener survey can be found in the supplemental materials.

In total, 19 participants were included in the study, with 9 identifying as men and 10 identifying as women. The mean age of users was 20 ($SD = 1.10$), with an age range from 18 to 23. Most of them identified as White ($n = 9$), followed by Asian ($n = 7$), Multiracial ($n = 2$), and one participant did not report their race. Regarding ethnicity, three participants self-identified as Latino/Hispanic, while the remaining 16 identified as not Latino/Hispanic. In terms of educational background, 6 participants had a high school diploma or equivalent, 11 had some college education or an associate degree, and two held a bachelor's degree. Generally, our participants scored none-to-moderate (i.e., scores within 0-14) across all symptoms, with an average PHQ-9 score of 5.57 ($SD = 4.36$), and an average GAD-7 score of 5.19 ($SD = 3.36$).

Study Procedure. Upon consent, the evaluation study consisted of two phases: 1) an onboarding and demonstration session to help users become familiar with the study procedures and CRBot, and 2) a two-week field study where users independently used CRBot in their everyday settings at their own pace and convenience, which allows users to chat with CRBot about potential unhelpful thinking in real-life scenarios. During this period, their cognitive restructuring dialogs were collected and stored in a private and secure database. At the end of the field study, participants completed an exit survey that included the System Usability Scale (SUS) [6], a Likert-scale measure of perceived effectiveness, and free-text questions on the tool's effectiveness, strengths, and limitations. The survey is provided in the supplementary material. Participants received a \$40 gift card as compensation for their time and effort.

Risk mitigation approaches. We adopted four measures to reduce potential risks:

- (1) User disclaimer: The chatbot prototype began each session by informing users that CRBot is not a replacement for professional mental health counseling, and clarifying both the scope and limitations of this research tool.
- (2) Suicidal ideation detection: We adapted a specialized prompt for scanning user inputs for extreme distress or self-harm references, building on previous results (e.g., 82% accuracy in detecting suicidal ideation) [5]. If suicidal ideation is detected, CRBot immediately presents additional support and crisis resources (e.g., helplines).
- (3) Monitoring dashboard: We developed a research dashboard that logs and updates user interactions. The team reviewed these logs multiple times daily and provided timely identification of high-risk conversations that warrant therapist follow-up.
- (4) Human oversight: Lastly, MHPs on our team performed regular checks on system outputs. If concerns or potential harms arise, they can intervene directly and provide relevant resources to users.

Data Analysis: We computed descriptive statistics to summarize dialog statistics and SUS score. For the qualitative data analysis of survey responses, we used the General Inductive Approach [44] to guide the thematic analysis of the open-ended responses. The first author read the free-text responses closely to gain an initial understanding of the concepts that emerged from the data, generated initial low-level codes, and organized them into preliminary high-level themes. These codes, together with their excerpts and candidate themes, were subsequently reviewed and discussed with the broader research team. Regular team discussions were held throughout the coding and clustering process to compare interpretations, resolve discrepancies, and refine the high-level themes.

3.3 Interview Study

Study Procedure. We conducted an expert review with four MHPs to gain deeper insights into how CRBot performed in real-world interactions. We met with each MHP multiple times (for an average of 3 times) over Zoom,

each lasting approximately an hour. During these sessions, they shared their screens and used the think-aloud method [49], articulating their thoughts and observations as they worked through each conversation log.

Each MHP was randomly assigned 24 conversation logs, approximately 18 pages per individual. We asked the MHPs to focus on several key areas: (1) the extent to which CRBOT's responses aligned with real-world therapeutic practices, (2) how well it guided users through cognitive restructuring, and (3) any ethical or safety concerns that emerged. When a misalignment or concern was flagged by an MHP, the same dialogue was assigned to multiple MHPs to cross-check their assessments. Inter-rater reliability (IRR) was measured through Krippendorff's Alpha α [17], and a substantial agreement was achieved ($\alpha = 0.75$). Divergences were addressed through consensus-building sessions [48], during which reviewers revisited either the transcript or relevant excerpts until alignment was reached.

Interview Data Analysis. Each think-aloud session was screen-recorded and transcribed verbatim. For the qualitative data analysis of cognitive restructuring dialogs, we used the General Inductive Approach [44] to guide the thematic analysis. The first author read the transcripts closely, generated low-level codes, and clustered related codes into preliminary high-level themes. These codes, together with their corresponding excerpts, and the emerging theme candidates were then reviewed and discussed with the broader research team. Throughout the coding and clustering process, the team engaged in regular discussions to compare interpretations, resolve coding discrepancies, and refine high-level themes.

4 Findings of User Study

We begin by presenting overall usage trends, the usability score, and perceived effectiveness. Next, we report emerging themes from qualitative analysis.

4.1 Engagement, Usability, and Perceived Effectiveness

As shown in Table 1, messages were exchanged in a near 1:1 ratio. Despite this overall balance, the standard deviations suggest considerable variation in engagement: some users held brief exchanges (as few as three total user messages), while others had lengthier conversations (up to 18 user messages). Likewise, the number of words in user inputs ranged from 16 to 373, indicating that some users provided short, concise statements, whereas others offered more elaborate reflections.

CR Dialogs (n=95)	Sum	Mean	SD	Min	Max
Number of Messages (User)	718	7.6	2.96	3	18
Number of Messages (Bot)	810	8.5	2.9	4	19
Number of Words (User)	8924	93.94	64.6	16	373
Number of Words (Bot)	23542	247.8	89.5	104	520

Table 1. Descriptive statistics for collected dialogs.

We used a system usability scale (SUS) to evaluate the overall usability of the tool. The average score is 71.62 ($SD = 11.21$), which indicates that the usability of CRBOT was seen positively (in general, a score above 68 is considered above average, according to the standards [21]).

Regarding the perceived effectiveness of our proposed method, participants were asked to provide ratings. The responses reflected a mixed distribution: most rated the approach as very effective ($n=7$), followed by slightly effective ($n=6$), moderately effective ($n=3$), and not effective at all ($n=2$), with one participant not responding. Overall, while the majority perceived benefits, the variation in responses highlights differing experiences with the approach.

4.2 Qualitative Findings

Three themes emerged from qualitative analysis of the exit survey: structure & guidance, empathy & validation, and psychoeducation.

Table 2. Codebook with themes, sub-themes, and example quotes from the user study (Step 2)

Themes	Subthemes	Example Quotes
Structure & Guidance	Step-by-step Guidance	“These [scaffolding] questions were effective in letting me analyze and think of the situation ... it let me take it one step at a time and let me understand the situation more rather than rushing my thinking.” (P10)
	Reflection & Self-awareness	“They effectively made me consider different ways my thoughts may be distorted instead of making me abstractly wonder ‘Are my thoughts distorted?’” (P5)
	Rigidity and Timing	“I think it could do with a bit less of the constant ‘do you think this is a distorted thought’ in every situation.” (P2)
	Over-simplified Questioning	“I think the question that asks ‘Do you think you successfully challenged your negative thoughts?’ could allow for you to say what part of your negative thoughts have been addressed and which ones still need further work” (P15)
Empathy & Validation	Compassionate & Non-judgmental stance	“The non-judgmental, empathetic approach made me more receptive to the tool’s guidance, since I didn’t feel ashamed or pressured.” (P16)
	Risk of Invalidation	“Sometimes it was so persistent, it felt like it was brushing aside my concerns or thoughts in favor of what it deemed to be reality.” (P1)
Psychoeducation	Labeling and Providing Explanations of Cognitive Distortions	“The ‘select distorted thought’ question was very helpful because I now have a label for what I think. I’ve found myself using these labels when I’m not using the tool to help me identify my distorted thinking.” (P8)
	Difficulty Applying Categories	“I think it was sometimes difficult to pick just one category” (P7)

4.2.1 Structure & Guidance. Participants provided both positive and negative feedback on how CRBot shaped their experiences, highlighting the trade-off between structure and flexibility.

An observable benefit of CRBot was its role in enabling **step-by-step guidance**. Several participants (n=8) noted that the stage-based flow simplified complex problems and created a sense of progress. For example, one reflected, “*They were effective in establishing a step-by-step path to ensuring I was able to identify and work through the distorted thought I had*” (P15). Similarly, another participant highlighted how adaptive scaffolding promoted clarity: “*These [scaffolding] questions were effective in letting me analyze and think of the situation ... it let me take it one step at a time and let me understand the situation more rather than rushing my thinking*” (P10).

Beyond providing structured flow, CRBot also fostered **reflection and self-awareness**. By utilizing techniques such as Socratic questioning, the tool encouraged users (n=10) to critically examine their own thoughts. One participant explained, “*They effectively made me consider different ways my thoughts may be distorted instead of*

making me abstractly wonder ‘Are my thoughts distorted?’” (P5). Another added, “As an overthinker, it really helped me come up with explanations for other people’s behavior that didn’t have to do with me which reduced the anxiety I was left feeling” (P15), suggesting that CRBOT helped them reframe anxious interpretations into more balanced perspectives.

Yet, not all experiences with CRBOT were positive. Several participants (n=8) criticized **rigidity and timing**, describing moments when scaffolding felt forced or premature. For instance, one participant noted, *“I think it could do with a bit less of the constant ‘do you think this is a distorted thought’ in every situation” (P2). Another reflected that, “I feel as though truly anxious people will never quite get to the point where they can honestly say they have fully challenged the thought, at least not as quickly as it was prompted in the tool” (P3). These reflections highlight the potential risks when pacing does not align with user readiness.*

Relatedly, some participants (n=5) critiqued the CRBOT’s reliance on **oversimplified questioning**, where complex thought patterns were reduced to yes/no choices for the scaffolding questions. One participant shared, *“They were ineffective when I felt that the options didn’t match my situation, because selecting ‘no’ just kind of sent me back to the beginning of the dialog, like the AI reset instead of pivoting” (P4). Another elaborated that, “I think the question that asks ‘Do you think you successfully challenged your negative thoughts?’ could allow for you to say what part of your negative thoughts have been addressed and which ones still need further work” (P15). These responses indicate that oversimplified questioning risks flattening nuanced reflections into procedural checkboxes, which may limit therapeutic engagement and hinder deeper thought exploration.*

4.2.2 Empathy & validation. Participants reflected on the system’s empathetic stance, praising its compassionate tone and human-like responsiveness while also cautioning that excessive persistence could lead to invalidation.

The majority of the participants (n=10) praised the system’s **compassionate and non-judgmental stance**. For instance, one participant shared, *“The non-judgmental, empathetic approach made me more receptive to the tool’s guidance, since I didn’t feel ashamed or pressured” (P16). Another noted, “I think the fact that it just felt like a person and would allow me to dump my feelings into it and it would actually respond with a human-like response.” (P12). These reflections suggest that empathy not only fostered comfort in disclosure but also increased receptivity to the therapeutic strategies offered.*

At the same time, two participants pointed to a **risk of invalidation** when persistence undermined empathy. One participant remarked, *“Sometimes it was so persistent, it felt like it was brushing aside my concerns or thoughts in favor of what it deemed to be reality” (P1). Another described the awkwardness of disclosing to an artificial system: “I think I felt a little awkward talking to an AI about my issues” (P7). Together, these accounts underscore the delicate balance between persistence and sensitivity. While empathy is critical in CR, over-insistence can produce alienation rather than validation.*

4.2.3 Psychoeducation. Participants provided both positive and negative feedback on the role of psychoeducation, highlighting the benefits of labeling and explanation while also noting challenges when categories felt restrictive.

Participants (n=12) emphasized that psychoeducation through **labeling and providing explanations of cognitive distortions** offered transferable knowledge beyond the immediate interaction. Labeling gave users a concrete vocabulary to recognize their own thinking patterns, both during and outside the tool. As one participant reflected, *“The ‘select distorted thought’ question was very helpful because I now have a label for what I think. I’ve found myself using these labels when I’m not using the tool to help me identify my distorted thinking” (P8). Another echoed this sentiment, noting, “The specific categories of distorted thoughts helped me identify where I can improve on in my thinking” (P11). Beyond naming distortions, participants valued the explanatory prompts that deepened reflection. For example, one remarked, “Labeling and explaining types of distorted thoughts and having me write a reframed thought after I finished talking to the chatbot were the most helpful” (P8). Another described how explanations strengthened clarity: “They make me be more specific about my problem” (P9). Collectively,*

these reflections suggest that psychoeducation equips participants with both the language and conceptual understanding needed to actively evaluate and refine their thoughts.

However, a minority of participants (n=2) described **difficulties with applying categories**. One remarked, “*I think it was sometimes difficult to pick just one category*” (P7), while another found that, “*The type of distorted thought was not very helpful*” (P13). These critiques highlight that while psychoeducation was broadly seen as beneficial, it must be designed to remain explanatory and adaptive, ensuring that educational content aids rather than constrains users’ self-understanding.

5 Findings of Interview Study

Our findings indicate that, while all MHPs acknowledged CRBot’s capacity to provide *protocol-adherent, natural, and in-depth conversations* and its capacity to *validate and normalize* users’ feelings, they also identified various concerns. These issues clustered around three themes: *the misuse of positive regard*, the indication of *power dynamics*, and the model’s *subjectivity and lack of contextual understanding*. Illustrative dialogue snippets for these themes are presented in Figure 4.

5.1 Protocol-Adherent, Natural, and In-Depth Conversations

MHPs emphasized several positive dimensions of CRBot’s interaction style, including its consistent adherence to cognitive restructuring steps, its fluid, and human-like conversational flow, and its ability to pose Socratic questions. All MHPs agreed that CRBot can adequately **follow the core phases of cognitive restructuring** (i.e., exploration, evaluation, and substitution). After reviewing multiple conversation logs, MHP1 commented,

“I can see that these conversations move into the cognitive restructuring framework.”

Similarly, MHP2 affirmed, “*I think overall, it’s doing a really good job of really honing in on the key components of cognitive restructuring.*” These comments illustrate that the scaffolding approach can effectively guide LLMs to adhere to CR protocols. Another strength pertained to CRBot’s ability to weave CR steps into a dialogue that felt **natural and conversational**. MHP2 noted,

“I think it does a good job of keeping it very conversational and mimicking what a real conversation would look like.”

Furthermore, MHP1 praised how the CRBot seamlessly integrated restructuring without overtly “announcing” each therapeutic step:

“The thing I like about this is you’re [CRBot] doing that without declaring, ‘Okay, now I’m going to restructure your thoughts...’ It’s much more conversational, much more natural.”

By delivering CR in a conversational manner, CRBot can help users feel less stigmatized or “*in therapy*,” potentially lowering barriers for those who might otherwise be hesitant to engage. Finally, all MHPs highlighted the CRBot’s **capacity to ask Socratic questions** [9], open-ended prompts aimed at helping users critically examine their thoughts. MHP2 observed,

“These responses are really good. They’re similar to the questions that I would ask... They make me explore the thought and challenge it on a deeper level.”

MHP2’s comment reflects a foundational element of CBT, where guided discovery encourages clients to generate their own insights rather than simply being told which thoughts are “right” or “wrong”. In addition, MHP3 added that,

“The questions are good at identifying evidence that supports [the distorted thought] and evidence against [the distorted thought].”

By prompting users to articulate both supportive and contradictory evidence for a particular belief, CRBot mirrored a therapist’s strategy of allowing clients to discover the validity (or lack thereof) in their own assumptions.

Themes	Subthemes	Example Quotes
Protocol-Adherent, Natural, and In-depth Conversations	Adherence to CR	“I can see that these conversations move into the cognitive restructuring framework.” (MHP1)
	Natural & conversational flow	“It does a good job of keeping it very conversational and mimicking what a real conversation would look like.” (MHP2)
	Capacity to pose Socratic questions	“These responses are really good. They’re similar to the questions that I would ask... They make me explore the thought and challenge it on a deeper level.” (MHP2)
Misuse of Positive Regard	Validation & empathy	“The bot is really good at giving validation, empathy, and normalization.” (MHP3)
	Toxic positivity	“That’s a great example” when they sort of even didn’t give an example... there’s a bit of a disconnect there.” (MHP3)
	Optimistic phrases overuse	“It doesn’t really tell the person what’s good, great, or awesome about it.” (MHP2)
Power Dynamics	Leading questions undermine autonomy	“Sometimes if [a leading question] is necessary, it’s just more effective if the client can get there first.” (MHP3)
	Evaluative language	“I also wonder if we could use a different word instead of ‘great’ to acknowledge the client’s achievement.” (MHP4)
	Advice-giving	“Advice giving is... at the top [of the therapeutic skill pyramid], where you use it with like discretion, a lot of context, and thought.” (MHP3)
Subjectivity & Context understanding	Misinterpretation of experience	“It’s leading them to think it was tough... But embarrassment is just that—embarrassment.” (MHP4)
	Misattribution of emotion	“They might have a healthy reaction—like it’s natural you’d be angry at this person... [but the bot tries] to challenge it.” (MHP3)
	Misinterpretation of subtle linguistic markers	“It seems like it’s [CRBot] taking ‘maybe’ to mean ‘yes’... when people say ‘maybe’ it’s often a little bit closer to a ‘no.’” (MHP2)
	Ignorance of session-wide patterns	“I’m not just looking at an individual response. I’m looking at the accumulation of responses, to see if there’s a pattern...” (MHP3)
	Unintended judgment	“Sometimes people can take the phrase ‘classic example’ as an insult.” (MHP1)

Table 3. Codebook with themes, sub-themes, and example quotes from MHPs (Step 3)

Such dialogue can foster deeper self-reflection and ownership of the therapeutic process—key contributors to sustained behavior change. Collectively, these findings suggest that CRBot can indeed capture important elements of CBT: from faithfully reproducing the “scaffolding” of cognitive restructuring to asking effective open-ended questions. Moreover, the natural feel of the conversation may increase user receptivity and reduce the stigma commonly associated with mental health interventions.

5.2 Misuse Positive Regards

Although all MHPs praised CRBot's aptitude for expressing validation, normalization, and empathy, they also cautioned against "toxic positivity" and the overuse of effusive praise. These observations highlight a delicate tension: while positive regard can foster hope and motivation, it may feel disingenuous or dismissive if delivered without sufficient nuance or context.

All MHPs commented favorably on CRBot's **capacity to validate users' emotions and normalize their experiences**. MHP3 commented, *"The bot is really good at giving validation, empathy, and normalization."* Such affirmative responses can be crucial for building trust and encouraging client self-disclosure. MHP1 further appreciated CRBot's linguistic variety:

"I like the fact that you're affirming of their experience in the initial response... in a different language, instead of just affirming the same way every time."

By avoiding overly formulaic expressions of empathy, CRBot came closer to replicating genuine human warmth. Despite these commendations, several MHPs worried about situations in which the chatbot's optimistic tone felt excessive or disconnected from the user's actual distress—commonly referred to as **"toxic positivity"** [7], which MHPs noted in four dialogues (4.2%). In one dialogue, the user provided a minimal, arguably inadequate example to challenge a distorted thought, yet CRBot responded, *"That's a great example!"* MHP3 noted the disconnect: *"That's a great example' when they sort of even didn't give an example... there's a bit of a disconnect there."* Similarly, MHP2 described a scenario in which the user's negative experiences were overshadowed by the bot's relentless positivity, recalling:

"It can feel almost invalidating... it doesn't undo the fact that these other people are being mean to me."

Such interactions may inadvertently minimize valid pain or frustration, mirroring broader concerns in psychotherapy that well-intentioned reassurance can sometimes trivialize or overlook genuine suffering.

Another related issue was CRBot's tendency to **overuse upbeat affirmations** like "That's great" or "That's wonderful," and such patterns were observed in six dialogues (6.3%). MHP2 emphasized that such blanket endorsements offer little insight: *"It doesn't really tell the person what's good, great, or awesome about it."* Interestingly, CRBot seemed to rely most heavily on these positive phrases when user engagement was minimal, sometimes reinforcing a sense of artificiality or "fakeness." Excessive positivity can erode the authenticity that person-centered therapies strive to cultivate [43], making the client doubt whether their more difficult emotions are being adequately heard.

MHP4 suggested that part of this disconnect stems from the chatbot's inability to convey tone as a human therapist might:

"I could say these very same words in a tone that the client would feel great without me telling them how great it is that they've done that."

This echoes broader findings that LLMs, despite their linguistic sophistication, cannot fully replicate paralinguistic cues (e.g., vocal intonation, facial expressions) crucial for nuanced emotional support.

5.3 Power Dynamics

Our analysis revealed that CRBot's conversational style sometimes created or reinforced power differentials, characterized by leading questions, evaluative or "definitive" praise, and advice-giving. While a certain level of power imbalance is inevitable in psychotherapy given the therapist's (or chatbot's) guiding role, as prior work suggests [32], our expert reviewers cautioned that unmoderated use of these language patterns might inadvertently reduce client autonomy or foster dependence. In what follows, we dissect how specific elements of

CRBot's output interacted with power dynamics, highlighting opportunities to recalibrate the tool's tone and prompts to better support users' sense of agency.

Our expert reviewers noted that while certain forms of guidance can help clients recognize and reframe negative thoughts, posing *leading questions* can inadvertently undermine autonomy. Such questions appeared in three dialogues (3.2%), typically when the bot steered users toward counter-evidence or conclusions that might otherwise have been reached on their own. For example, when CRBot provided a direct example of evidence against a user's distorted thought, MHP3 commented,

"Sometimes it [a leading question] is necessary, it's just more effective if the client can get there first."

This comment echoes a key principle in many therapeutic modalities: clients are more likely to internalize and sustain insights that they arrive at independently [35]. By preemptively supplying counter-evidence or steering users to a "correct" response, the tool risks bypassing users' own reflective processes. A related concern emerged when CRBot ended a question with "right?" prompting MHP2 to observe,

"So I would avoid questions that end in right, because people who have a lot of anxiety a lot of times if you pose a question and you end it in right, whether they believe it or not, they're going to agree... they want to please whoever it is."

Indeed, highly anxious individuals are more likely to exhibit conformity toward others, as prior work has suggested [58]; in this case, the user may conform to perceived "expert" opinions in an effort to avoid conflict or judgment. From a clinical perspective, such patterns can stifle genuine self-exploration, limiting the user's sense of ownership over their therapeutic journey. Taken together, these observations highlight the delicate balance between offering supportive prompts and inadvertently overdirecting users. While leading questions can quickly scaffold a session (e.g., helping a client identify and challenge a specific cognitive distortion), they should be used sparingly and skillfully to maintain a collaborative stance and promote user self-efficacy.

A second concern centered on the excessive usage of phrases such as "That's great" or "That's wonderful," which, according to the MHPs, risked shifting the therapist-client dynamic from collaborative exploration to performance assessment. For example, MHP4 remarked,

"I also wonder if we could use a different word instead of 'great' to acknowledge the client's achievement. What if they didn't achieve? Does this mean that they're not great anymore?"

MHP4's comment points to how highly evaluative language can become counterproductive if a user interprets it as a definitive judgment of their progress or self-worth. MHP4 further explained how definitive statements can exacerbate underlying power differential:

"Using words like this sometimes could lead to the user working so hard to be great, because the person who they're coming for help from has told them to be great. So if I get 'great' one time, what if I don't get it the second time?"

Similarly, MHP3 noted, *"I also generally don't label things as good or bad, or great or not great. I label them as helpful or unhelpful,"* indicating a general preference for neutral statements. Altogether, these observations echo broader therapy guidelines that favor process-oriented over evaluative language [46]. Subtly reframing "That's great!" into "It sounds like you found something that works for you" helps foster self-reflection without amplifying external validation.

Under the broader issue of power dynamics, another salient subtheme emerged around *advice giving*, which was identified in six dialogues (6.3%), with MHPs warning that even seemingly innocuous suggestions can magnify the CRBot's authority. In traditional psychotherapy, clinicians are already positioned as experts, and offering advice can increase the power differential, especially if clients feel compelled to comply simply due to the therapist's status. When advice is offered by an LLM, this effect can become further magnified by the model's perceived "objectivity" or infallibility.

In our study, expert reviewers pointed out that a certain amount of direction is warranted in therapy (e.g., encouraging skill practice), but unprompted or overly prescriptive advice risks undermining user autonomy and may lead to unintended consequences. In most cases, CRBot’s suggestions merely reinforced strategies already covered, which the MHPs viewed as appropriate. As MHP2 noted,

“So I think in this situation it’s fine, because you’re just encouraging them to continue practicing the skill that they’re learning... But I would not want AI to give users advice on other things like, ‘Oh, you should try this,’ or ‘You should say this in this conversation.’”

Here, MHP2 highlights the tension between supporting therapeutic techniques, such as prompting users to do cognitive restructuring, and extending into broader, potentially uncontextualized advice. A real-world example involved a user who forgot a friend’s birthday; when CRBot suggested explaining their forgetfulness, the user retorted, *“It is rude,”* indicating the advice did not fit the situation. From a psychotherapy standpoint, advice is most helpful when a therapeutic relationship has been established and extensive exploration and insight have occurred, which requires accurate clinical judgment [12]. As MHP3 emphasized:

“Advice giving is... at the top [of the therapeutic skill pyramid], where you use it with discretion, a lot of context, and thought... Worst case scenario: what if this person the client’s talking about is physically abusive?... we’re saying, ‘Hey, you should go talk to them...’”

5.4 Subjectivity and Lack of Context Understanding

A recurring critique raised by the MHPs concerned the chatbot’s tendency to oversimplify or misread user experiences. Although our experts generally acknowledged CRBot’s capacity to summarize and reflect on users’ concerns, they also encountered multiple cases of misinterpretation, misunderstanding of implicit cues, overlooking session-wide behavior, and unintentionally judgmental phrasing.

In psychotherapy, accurately capturing and reflecting a client’s emotional state is pivotal for establishing rapport and fostering therapeutic alliance [31]. By mirroring the client’s language, therapists convey understanding, validate the client’s perspective, and invite further reflection. In contrast, our MHPs noted four dialogs (4.2%) where CRBot inadvertently **distorted users’ stated experiences**, potentially undermining this reflective process.

A telling example occurred when a user described feeling “embarrassed,” but CRBot summarized the experience as “tough.” MHP4 mentioned,

“It’s leading them to think it was tough... Now I’m putting myself at the center of that conversation to say embarrassment for me is tough. But embarrassment is just that—embarrassment.”

Such a small shift in wording may appear innocuous, yet it highlights how a seemingly “synonymous” reframe can redirect users away from their own nuanced feelings. In another case, a user felt “stressed” about not completing enough work, yet CRBot assumed a sole source of stress, unsatisfied in work progress. MHP1 questioned,

“Are you nervous because you’ve not done any work, or are you nervous because you’re not believing your work is enough?”

In this case, the stress might also stem from the lack of completed work itself, leading to distinct underlying irrational thoughts and, consequently, different therapeutic strategies. These reflections underscore the need for careful, iterative prompt design and potential human oversight when employing LLMs in mental health interventions.

Despite our efforts to mitigate pathologization by including a few-shot example, such instances still occurred, highlighting the need for more sophisticated guardrails. The concern, observed in three dialogs (3.2%), arose when CRBot **attributed typical emotional responses, such as anger, to maladaptive beliefs**, MHP3 explained,

“They might have a healthy reaction—like it’s natural you’d be angry at this person... [but the bot tries] to challenge it.”

In many therapeutic approaches, experiencing anger can be both appropriate and adaptive—for example, signaling the need to establish boundaries or acknowledge unmet needs [3, 11]. Pathologizing such emotions or treating them as inherently “negative” risks invalidating a client’s feelings and might inadvertently dissuade them from engaging in self-reflection. Another common theme was the tendency to **misinterpret subtle linguistic markers**. This occurred in two dialogues (2.1%), where words and phrases such as “maybe” or “I guess” were treated as agreement rather than ambivalence, as MHP2 noted:

“It seems like it’s [CRBOT] taking ‘maybe’ to mean ‘yes’. However, a lot of times when people say ‘maybe’ it’s often a little bit closer to a ‘no’ than a ‘yes’”

In psychotherapy, these markers often signal ambivalence, uncertainty, or emotional conflict [18]. However, CRBOT frequently treated these tentative expressions as if they were genuine agreements, plowing ahead with encouragement or a new line of questioning. Such behavior contrasts sharply with the human therapist’s approach. As MHP3 stated,

“Whenever I give them the rationale for the healthier thought before they’re ready for it. They’ll kind of just tell me, like, sure, maybe, and then that’s when I usually pause and explore a little bit more of their uncertainty.”

This difference can partially be attributed to the limitations of text-based communication in capturing nonverbal cues. In a live therapy session, a clinician would likely sense the user’s ambivalence through vocal tone, facial expression, or body language—cues that are inaccessible to text-based LLMs. By overlooking these nuances, CRBOT can inadvertently rush clients through critical points of self-reflection or emotional processing, potentially missing an opportunity to address deeper reservations or unspoken discomfort.

MHPs also mentioned CRBOT for **treating each user turn in isolation**, rather than considering broader patterns across the entire session. This limitation was flagged in two dialogues (2.1%), where the user responded with consecutive short, disinterested phrases, CRBOT continued pushing cognitive restructuring steps. MHP3 contrasted this with a more human response:

“I’m not just looking at an individual response. I’m looking at the accumulation of responses, to see if there’s a pattern... this person doesn’t want to engage.”

For a human therapist, repeatedly short or disinterested replies may indicate underlying resistance, fatigue, or even deeper emotional blocks. Therapists might then shift gears—e.g., inquire about external stressors, try a different intervention, or explicitly acknowledge the client’s reluctance—instead of persisting along the original therapeutic track [56]. In contrast, CRBOT mechanically continues cognitive restructuring, failing to register the user’s withdrawal or frustration, which risks eroding rapport and halting therapeutic progress.

A further concern was CRBOT’s occasional use of language that, while not overtly critical, might still feel **judgmental** to users. Such instances were observed in four dialogs (4.2%), where MHPs pointed to phrases like “classic example” or invitations to produce a “more positive thought” as subtle cues that could imply the user’s situation or perspective falls short of an ideal. As MHP1 observed,

“Sometimes people can take the phrase ‘classic example’ as an insult...”

Meanwhile, MHP2 noted,

“When you say ‘a more positive thought,’ it introduces that evaluative element of right and wrong.”

Such phrasing risks introducing a moral or normative undertone, which may inadvertently pressure users to align with an externally imposed benchmark rather than explore their emotions freely. In many counseling traditions, especially person-centered therapy, clinicians are taught to maintain a nonjudgmental stance—using neutral, open-ended language (e.g., “Could you think of a thought that feels less anxiety-provoking?”) instead of framing the alternative thought as “better” or “more correct”. This stance fosters an atmosphere of psychological safety, encouraging clients to express themselves without fear of disapproval. When a chatbot implicitly casts

certain emotions or cognition as suboptimal, it can disrupt that sense of safety by conveying, however subtly, that certain thoughts or feelings are “wrong.”

Taken together, these examples, ranging from the misinterpretation of user sentiments to unintended judgment, illustrate CRBot’s limitations in nuanced contextual understanding and its tendency toward subjectivity, thereby limiting the richness and precision of therapeutic engagement.

6 Discussion

Building on our findings, we discuss implications for advancing LLM-enabled CR across four dimensions: (1) the trade-offs between structured guidance, psychoeducation, and flexibility in adaptive scaffolding (see Section 6.1), (2) the expression and recognition of emotion, including positive regard and contextual sensitivity (see Section 6.2), (3) power dynamics and user autonomy (see Section 6.3), and (4) moving toward safer LLM-based CR through rethinking evaluation strategies and strengthening human oversight (see Section 6.4). One important caveat is that our findings are drawn from a small ($n=19$), young (mean age=20), and non-clinical sample with none-to-moderate levels of distress. Therefore, implications discussed below should be treated with caution, as safety considerations may differ substantially for clinical populations or higher-risk users with more acute mental health needs.

6.1 Adaptive Scaffolding: Trade-off Between Structured Guidance, Psychoeducation, and Flexibility

Users and MHPs largely converged in recognizing the value of CRBot’s structured approach, but both also identified rigidity as a limitation. From the user side, scaffolding questions felt forced and premature, with pacing that did not align with their readiness. Several users noted that “yes/no” scaffolding questions flattened nuanced reflections into binary questioning, limiting engagement and hindering deeper exploration. Particularly, we noticed that in the second stage (“challenge distorted thoughts”), the CRBot could not reliably determine whether users had fully engaged with their beliefs before moving on, resulting in premature or inappropriate transitions. Similarly, MHPs raised a complementary concern: that the system continued predefined CR steps regardless of user disengagement, failing to react when consecutive short and disinterested responses were presented.

Although psychoeducation was broadly valued, its implementation also introduced drawbacks. Some users reported difficulty in selecting a single distortion type, noting that categories sometimes felt restrictive or failed to capture the complexity of their thought patterns. A likely explanation is the simultaneous presence of multiple distortions, which the system’s one-choice design did not accommodate. Without such flexibility, psychoeducation risks oversimplifying users’ complex thought patterns and limiting their ability to accurately recognize how different distortions interact in their own thinking.

Taken together, these findings suggest three design implications. First, future CR systems should include richer psychoeducation features that allow multiple distortions to be selected. Future research could explore how to manage dialogue states when multiple distortions coexist—for instance, by determining the priority of distortions or enabling branching conversations that address them sequentially. Second, future work should improve detection accuracy at transition points, such as determining whether a user has genuinely challenged a thought. This may be achieved through more sophisticated prompt-engineering methods (e.g., chain-of-thought prompting [52]) or hybrid approaches that combine lightweight classifiers with LLMs’ generative capability (e.g., classifying user utterances and feeding the classification results into the LLM [25]), with classifiers specifically trained for CR-specific states. Third, adaptive scaffolding frameworks could benefit from broader modalities to enhance the expressiveness of user interactions. For example, rather than relying on binary questions such as “Do you think you have successfully challenged the distorted thought?” (as in our CRBot design), future studies could adopt more flexible phrasing (e.g., “How do you feel about your progress in challenging this thought?”) coupled with a Likert-style scale ranging from “Not confident at all” to “Strongly confident.” Such adaptable forms

may enable users to articulate their progress with greater nuance and provide the system with more reliable signals of their therapeutic engagement.

6.2 Expressing and Recognizing Emotion: Positive Regard and Contextual Sensitivity

While both users and MHPs acknowledged CRBot's empathetic tone as a strength, MHPs additionally identified nuanced but substantial risks that were not immediately apparent to users. Users frequently praised CRBot's non-judgmental stance, with the majority highlighting it as one of the tool's most valued qualities. MHPs, while affirming CRBot's capacity for validation and normalization, raised concerns about several risky patterns. Specifically, they identified instances of toxic positivity, where positivity was applied inappropriately and disconnected from the user's actual response, inadvertently invalidating genuine distress. MHPs also noted that the excessive use of phrases like "That's great!" risked eroding the authenticity that therapy strives to cultivate.

Beyond expressing empathy, MHPs also identified limitations in CRBot's ability to accurately recognize users' emotional states. CRBot showed limited capacity to interpret nuanced emotions, complex experiences, and linguistic markers of ambivalence, for instance, treating "maybe" as agreement rather than uncertainty. Moreover, CRBot tended to treat each user turn in isolation rather than recognizing broader conversational patterns across the session, an ability human therapists rely on to detect disengagement, resistance, or withdrawal. Collectively, these limitations suggest that CRBot's relational challenges extend beyond tone and word choice to a more fundamental difficulty in reading the emotional nuances of conversations.

As such, we provide two recommendations for future design and research. First, recognizing empathy as a nuanced and multi-dimensional construct, future work can systematically examine patterns of LLM-generated empathy and develop safeguards, such as classifiers that detect potentially hollow and toxic positivity, to protect users from invalidation. One approach is to use prompt engineering to distinguish contextually grounded affirmations (e.g., "It sounds like you're starting to see this differently") from generic blanket endorsements (e.g., "That's great!"), instructing the system to always link affirmative language to the specific insight or effort the user has demonstrated. Second, studies on LLM-based CR should explore integrating modalities beyond text, building on prior work that combines LLMs with audio [55], video [26], wearable sensors [27], or virtual reality [54]. Such multi-modal integration could provide richer signals of user affect (e.g., detecting ambivalence, sustained disengagement, or heightened distress). By capturing session-wide emotional trajectories and nonverbal cues, multi-modal LLM systems may better align with the relational depth of human therapy, enhancing both empathy delivery and therapeutic efficacy.

6.3 Navigating Power Dynamics: Advice-Giving, Authority, and User Autonomy

Power imbalance was not a primary concern among users, who generally experienced CRBot's guidance as helpful. MHPs, however, identified several language patterns as problematic, including leading questions, evaluative language, and advice-giving. Of these, advice-giving poses the most direct concern, as it requires precise clinical judgment and extensive contextual knowledge that LLMs cannot reliably satisfy [20, 24]. This risk is compounded by evidence that users tend to over-rely on algorithmic outputs: foundational work on the Machine Heuristic [42], Automation Bias [10], and Algorithm Appreciation [23] establishes that algorithmic sources carry an implicit authority that is difficult to resist, and more recent empirical work suggests this pattern of uncritical reliance extends to LLM-based mental health intervention where users may be especially susceptible to authoritative guidance [14, 38]. Evaluative language and leading questions compound the power dynamics in subtler ways, potentially shifting the interaction from collaborative exploration toward performance evaluation and prompting users to conform to the system's framing rather than arrive at insights independently, both of which undermine the user autonomy that effective CR fundamentally depends on.

To mitigate risks, future designs could consider several practical approaches. First, LLM-based CR systems should implement multiple layers of guardrails around advice-giving. At the most restrictive level, certain categories of advice (e.g., medical or interpersonal advice) should be explicitly excluded from the system’s output, as these carry high potential for harm when delivered without knowledge of the user’s full situation. For other forms of advice that remain within the scope of CR practice, attaching explicit disclaimers at the point of delivery (e.g., “This is one possibility based on what you have shared, but it may not fit your situation, please trust your own judgment”) helps remind users of the system’s limitations. Second, to reduce the risks of evaluative language and leading questions, future work could consider aligning LLM responses with therapeutic standards through reinforcement learning from human feedback (RLHF) [30], using expert therapist ratings to train the model toward more autonomy-preserving language.

6.4 Toward Safer LLM-Based CR: Beyond Protocol Evaluation and Human Oversight

Beyond the specific design recommendations, our study has broader implications for how LLM-based CR tools should be evaluated. First, although MHPs did not identify immediate ethical concerns in this study, they flagged a recurring pattern of behaviors across multiple dimensions: the misuse of positive regard, where positivity risked trivializing users’ genuine distress; contextual understanding gaps that led to misattribution of emotions and misinterpretation of subtle linguistic cues; and power imbalances introduced through evaluative language, leading questions, and advice-giving. Taken together, these behaviors illustrate an important implication: protocol adherence does not imply safe or therapeutically sound interaction. An LLM can faithfully follow the structural steps of CR while simultaneously invalidating users’ emotional experiences, exerting improper influence over their responses, or missing the contextual nuances that determine whether a response is appropriate. Designers of LLM-based CR tools, therefore, need to treat protocol adherence and therapeutic soundness as distinct and equally weighted design dimensions. Future evaluations should assess not only whether the system completes each CR step correctly, but also whether doing so appropriately expresses positive regard, accurately recognizes and responds to the user’s emotional state, demonstrates sufficient contextual understanding, and preserves the user’s sense of agency throughout the interaction.

Second, future studies should capture both subjective user experience and expert clinical assessment, rather than relying on either as the primary feasibility signal. As our findings demonstrate, users and MHPs can diverge substantially in their evaluation of the same interactions: features that users experience as supportive and validating may simultaneously carry clinical risks that only trained professionals can identify. Evaluation pipelines that incorporate ongoing expert review of interaction logs, analogous to the MHP review conducted in this study, would provide a more comprehensive and clinically grounded basis for assessing system safety.

Finally, human oversight remains the necessary safeguard. Given the nuanced but significant challenges identified in this study, LLM-based CR is not yet mature enough to operate reliably without human supervision. Human-in-the-loop strategies, such as regular MHP review of system outputs and real-time monitoring dashboards, are therefore essential to ensure that edge cases, emerging risks, and subtle harms that evade automated detection can be identified and addressed in a timely manner. Future evaluations of LLM-based CR systems should therefore incorporate human oversight, such as ongoing expert review of interaction logs, as a standard component of the evaluation pipeline.

7 Limitations and Future Work

While our study offers valuable insights into the potential and pitfalls of LLM-based psychotherapy, several constraints must be acknowledged. First, to mitigate risks, we limited participants to individuals without severe mental health concerns, excluding those with more acute needs who might use LLM-based tools differently. Yet, it is important to design and deploy LLM-powered systems safely before running a large-scale human-subjects

study with broader populations. Second, although we asked MHPs to annotate potential concerns and instances of misalignment with real-world therapy practice, we did not ask them to provide corrected responses. Such corrections could serve as a valuable resource for developing future CR systems, and we plan to incorporate this step in subsequent studies. Finally, although we aimed to capture diverse therapist perspectives, our panel's professional orientations and cultural backgrounds might still be limited, potentially skewing interpretations of the chatbot's responses. As such, future work can engage therapists from varied theoretical and cultural contexts to offer a more comprehensive understanding of how such LLMs perform across different clinical backgrounds and demographics.

8 Conclusion

In this work, we presented the evaluation of CRBot, an LLM-based cognitive restructuring system, by gathering user feedback and inviting mental health professionals to assess its performance on real-world transcripts. Our findings show that while the system can scaffold the core steps of cognitive restructuring, support self-reflection, and convey empathetic and non-judgmental responses, it also demonstrated limitations, including rigidity, oversimplified questioning, misuse of positive regard, pathologization of appropriate emotions, and difficulties in interpreting nuanced context. To address these challenges, we recommend that future work enhance adaptive scaffolding with greater flexibility; develop safeguards for contextually grounded empathy expression; implement guardrails around advice-giving; and move toward evaluation that treats protocol adherence and therapeutic soundness as distinct dimensions, with human oversight as a necessary component. We hope this work provides insights for designing safe, effective, and contextually sensitive LLM-enabled CR systems that better align with the realities of therapeutic practice.

References

- [1] Aaron T Beck. 1979. *Cognitive therapy and the emotional disorders*. Penguin.
- [2] Judith S Beck. 2020. *Cognitive behavior therapy: Basics and beyond*. Guilford Publications.
- [3] Olga V Berkout, Diana Tinsley, and Maureen K Flynn. 2019. A review of anger, hostility, and aggression from an ACT perspective. *Journal of contextual behavioral science* 11 (2019), 34–43.
- [4] Priya Bhatt, Amanrose Sethi, Vaibhav Tasgaonkar, Jugal Shroff, Isha Pendharkar, Aditya Desai, Pratyush Sinha, Aditya Deshpande, Gargi Joshi, Anil Rahate, et al. 2023. Machine learning for cognitive behavioral analysis: datasets, methods, paradigms, and research directions. *Brain informatics* 10, 1 (2023), 18. <https://doi.org/10.1186/s40708-023-00196-6>
- [5] Pierre-William Breau. 2023. Low-resource suicide ideation and depression detection with multitask learning and large language models. (2023). <https://hdl.handle.net/1866/32349>
- [6] John Brooke et al. 1996. SUS-A quick and dirty usability scale. *Usability evaluation in industry* 189, 194 (1996), 4–7.
- [7] Laura E Captari, Steven J Sandage, Richard A Vandiver, Peter J Jankowski, and Joshua N Hook. 2023. Integrating positive psychology, religion/spirituality, and a virtue focus within culturally responsive mental healthcare. *Handbook of positive psychology, religion, and spirituality* (2023), 413.
- [8] David A Clark. 2013. Cognitive restructuring. *The Wiley handbook of cognitive behavioral therapy* (2013), 1–22. <https://doi.org/10.1002/9781118528563.wbcbt02>
- [9] Gavin I Clark and Sarah J Egan. 2015. The Socratic method in cognitive behavioural therapy: a narrative review. *Cognitive Therapy and Research* 39 (2015), 863–879.
- [10] Mary Cummings. [n. d.]. *Automation Bias in Intelligent Time Critical Decision Support Systems*. <https://doi.org/10.2514/6.2004-6313> arXiv:<https://arc.aiaa.org/doi/pdf/10.2514/6.2004-6313>
- [11] Jerry L Deffenbacher. 2011. Cognitive-behavioral conceptualization and treatment of anger. *Cognitive and Behavioral Practice* 18, 2 (2011), 212–221.
- [12] Changming Duan, Sarah Knox, and Clara E Hill. 2018. Advice giving in psychotherapy. *The Oxford handbook of advice* (2018), 175–195.
- [13] Zhijun Guo, Alvina Lai, Johan Thygesen, Joseph Farrington, Thomas Keen, and Kezhi Li. 2024. Large Language Model for Mental Health: A Systematic Review (Preprint). (Feb. 2024). <https://doi.org/10.2196/preprints.57400>
- [14] Zainab Iftikhar, Amy Xiao, Sean Ransom, Jeff Huang, and Harini Suresh. 2025. How LLM counselors violate ethical standards in mental health practice: A practitioner-informed framework. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, Vol. 8. 1311–1323.

- [15] Lana Kambaitz-Illankovic, Uma Rzyeva, Laura Völkel, Julian Wenzel, Johanna Weiske, Frank Jessen, Ulrich Reininghaus, Peter J Uhlhaas, Mario Alvarez-Jimenez, and Joseph Kambaitz. 2022. A systematic review of digital and face-to-face cognitive behavioral therapy for depression. *NPJ Digital Medicine* 5, 1 (2022), 144.
- [16] Mina J Kian, Mingyu Zong, Katrin Fischer, Abhyuday Singh, Anna-Maria Velentza, Pau Sang, Shriya Upadhyay, Anika Gupta, Misha A Faruki, Wallace Browning, et al. 2024. Can an LLM-powered socially assistive robot effectively and safely deliver cognitive behavioral therapy? A study with university students. *arXiv preprint arXiv:2402.17937* (2024). <https://doi.org/10.48550/arXiv.2402.17937>
- [17] Klaus Krippendorff. 2011. Computing Krippendorff's alpha-reliability. (2011).
- [18] Anton O Kris. 1984. The conflicts of ambivalence. *The Psychoanalytic Study of the Child* 39, 1 (1984), 213–234.
- [19] Kurt Kroenke, Robert L Spitzer, and Janet BW Williams. 2001. The PHQ-9: validity of a brief depression severity measure. *Journal of general internal medicine* 16, 9 (2001), 606–613. <https://doi.org/10.1046/j.1525-1497.2001.016009606.x>
- [20] Hannah R Lawrence, Renee A Schneider, Susan B Rubin, Maja J Matarić, Daniel J McDuff, and Megan Jones Bell. 2024. The opportunities and risks of large language models in mental health. *JMIR Mental Health* 11, 1 (2024), e59479.
- [21] James R Lewis. 2018. Measuring perceived usability: The CSUQ, SUS, and UMUX. *International Journal of Human-Computer Interaction* 34, 12 (2018), 1148–1156. <https://doi.org/10.1080/10447318.2017.1418805>
- [22] Joanna Zun Li, Alina Herderich, and Amit Goldenberg. 2024. Skill but not effort drive gpt overperformance over humans in cognitive reframing of negative scenarios. (2024).
- [23] Jennifer M Logg, Julia A Minson, and Don A Moore. 2019. Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational behavior and human decision processes* 151 (2019), 90–103.
- [24] Rakesh K Maurya, Steven Montesinos, Mikhail Bogomaz, and Amanda C DeDiego. 2025. Assessing the use of ChatGPT as a psychoeducational tool for mental health practice. *Counselling and Psychotherapy Research* 25, 1 (2025), e12759.
- [25] Selina Meyer and David Elswiler. 2025. LLM-based conversational agents for behaviour change support: A randomised controlled trial examining efficacy, safety, and the role of user behaviour. *International Journal of Human-Computer Studies* 200 (2025), 103514.
- [26] Palash Moon and Pushpak Bhattacharyya. 2024. We Care: Multimodal Depression Detection and Knowledge Infused Mental Health Therapeutic Response Generation. In *Proceedings of the 21st International Conference on Natural Language Processing (ICON)*, Sobha Lalitha Devi and Karunesh Arora (Eds.). NLP Association of India (NLPAl), AU-KBC Research Centre, Chennai, India, 296–310. <https://aclanthology.org/2024.icon-1.35/>
- [27] Sameer Neupane, Poorvesh Dongre, Denis Gracanin, and Santosh Kumar. 2025. Wearable Meets LLM for Stress Management: A Duoethnographic Study Integrating Wearable-Triggered Stressors and LLM Chatbots for Personalized Interventions. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Association for Computing Machinery, New York, NY, USA, Article 588, 8 pages. <https://doi.org/10.1145/3706599.3720197>
- [28] Jingping Nie, Hanya (Vera) Shao, Yuang Fan, Qijia Shao, Haoxuan You, Matthias Preindl, and Xiaofan Jiang. 2025. LLM-based Conversational AI Therapist for Daily Functioning Screening and Psychotherapeutic Intervention via Everyday Smart Devices. *ACM Trans. Comput. Healthcare* (Jan. 2025). <https://doi.org/10.1145/3712299> Just Accepted.
- [29] John C Norcross and Michael J Lambert. 2018. Psychotherapy relationships that work III. *Psychotherapy* 55, 4 (2018), 303.
- [30] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. 2022. Training language models to follow instructions with human feedback. *Advances in neural information processing systems* 35 (2022), 27730–27744.
- [31] Rafael Zambelli Pinto, Manuela L Ferreira, Vinicius C Oliveira, Marcia R Franco, Roger Adams, Christopher G Maher, and Paulo H Ferreira. 2012. Patient-centred communication is associated with positive therapeutic alliance: a systematic review. *Journal of physiotherapy* 58, 2 (2012), 77–87.
- [32] Kenneth S Pope and Melba JT Vasquez. 2016. *Ethics in psychotherapy and counseling: A practical guide*. John Wiley & Sons.
- [33] Paolo Raile. 2024. The usefulness of ChatGPT for psychotherapists and patients. *Humanities and Social Sciences Communications* 11, 1 (2024), 1–8.
- [34] Lina M. Rojas-Barahona, Bo-Hsiang Tseng, Yinpei Dai, Clare Mansfield, Osman Ramadan, Stefan Ultes, Michael Crawford, and Milica Gašić. 2018. Deep learning for language understanding of mental health concepts derived from Cognitive Behavioural Therapy. In *Proceedings of the Ninth International Workshop on Health Text Mining and Information Analysis*, Alberto Lavelli, Anne-Lyse Minard, and Fabio Rinaldi (Eds.). Association for Computational Linguistics, Brussels, Belgium, 44–54. <https://doi.org/10.18653/v1/W18-5606>
- [35] Richard M Ryan, Martin F Lynch, Maarten Vansteenkiste, and Edward L Deci. 2011. Motivation and autonomy in counseling, psychotherapy, and behavior change: A look at theory and practice 1/7. *The Counseling Psychologist* 39, 2 (2011), 193–260.
- [36] Ashish Sharma, Kevin Rushton, Inna Lin, David Wadden, Khendra Lucas, Adam Miner, Theresa Nguyen, and Tim Althoff. 2023. Cognitive Reframing of Negative Thoughts through Human-Language Model Interaction. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki (Eds.). Association for Computational Linguistics, Toronto, Canada, 9977–10000. <https://doi.org/10.18653/v1/2023.acl-long.555>
- [37] Ashish Sharma, Kevin Rushton, Inna Wanyin Lin, Theresa Nguyen, and Tim Althoff. 2024. Facilitating Self-Guided Mental Health Interventions Through Human-Language Model Interaction: A Case Study of Cognitive Restructuring. In *Proceedings of the 2024 CHI*

- Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 700, 29 pages. <https://doi.org/10.1145/3613904.3642761>
- [38] Inhwa Song, Sachin R Pendse, Neha Kumar, and Munmun De Choudhury. 2025. The typing cure: Experiences with large language model chatbots for mental health support. *Proceedings of the ACM on Human-Computer Interaction* 9, 7 (2025), 1–29.
- [39] Robert L Spitzer, Kurt Kroenke, Janet BW Williams, and Bernd Löwe. 2006. A brief measure for assessing generalized anxiety disorder: the GAD-7. *Archives of internal medicine* 166, 10 (2006), 1092–1097. <https://doi.org/10.1001/archinte.166.10.1092>
- [40] Elizabeth C Stade, Shannon Wiltsey Stirman, Lyle H Ungar, Cody L Boland, H Andrew Schwartz, David B Yaden, Joao Sedoc, Robert J DeRubeis, Robb Willer, and Johannes C Eichstaedt. 2024. Large language models could change the future of behavioral healthcare: A proposal for responsible development and evaluation. *npj Mental Health Research*, 3 (1): 1–12, April 2024. <https://doi.org/10.1038/s44184-024-00056-z>
- [41] Xin Sun, Jan de Wit, Zhuying Li, Jiahuan Pei, Abdallah El Ali, and Jos A Bosch. 2024. Script-Strategy Aligned Generation: Aligning LLMs with Expert-Crafted Dialogue Scripts and Therapeutic Strategies for Psychotherapy. *arXiv preprint arXiv:2411.06723* (2024).
- [42] S. Shyam Sundar and Jinyoung Kim. 2019. Machine Heuristic: When We Trust Computers More than Humans with Our Personal Information. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–9. <https://doi.org/10.1145/3290605.3300768>
- [43] Jessica Y Suzuki. 2018. *A Qualitative Investigation of Psychotherapy Clients' Perceptions of Positive Regard*. Columbia University.
- [44] David R Thomas. 2006. A general inductive approach for analyzing qualitative evaluation data. *American journal of evaluation* 27, 2 (2006), 237–246. <https://doi.org/10.1177/109821400528374>
- [45] Fouad Trad and Ali Chehab. 2024. Prompt engineering or fine-tuning? a case study on phishing detection with large language models. *Machine Learning and Knowledge Extraction* 6, 1 (2024), 367–384. <https://doi.org/10.3390/make6010018>
- [46] Charles B Truax. 1970. Therapist's evaluative statements and patient outcome in psychotherapy. *Journal of Clinical Psychology* 26, 4 (1970).
- [47] Conal Twomey, Gary O'Reilly, and Michael Byrne. 2015. Effectiveness of cognitive behavioural therapy for anxiety and depression in primary care: a meta-analysis. *Family practice* 32, 1 (2015), 3–15.
- [48] Marleen Van de Kerkhof. 2006. Making a difference: on the constraints of consensus building and the relevance of deliberation in stakeholder dialogues. *Policy Sciences* 39, 3 (2006), 279–299.
- [49] Maarten Van Someren, Yvonne F Barnard, and J Sandberg. 1994. The think aloud method: a practical approach to modelling cognitive. *London: Academic Press* 11, 6 (1994).
- [50] Milton L Wainberg, Pamela Scorza, James M Shultz, Liat Helpman, Jennifer J Mootz, Karen A Johnson, Yuval Neria, Jean-Marie E Bradford, Maria A Oquendo, and Melissa R Arbuckle. 2017. Challenges and opportunities in global mental health: a research-to-practice perspective. *Current psychiatry reports* 19 (2017), 1–10.
- [51] Xiaomeng Wang, Dharmendra Sharma, and Dinesh Kumar. 2024. Cognitive Reframing via Large Language Models for Enhanced Linguistic Attributes. In *The Second Tiny Papers Track at ICLR 2024*. <https://openreview.net/forum?id=Ilus8CSwsU>
- [52] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Proceedings of the 36th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) (NIPS '22). Curran Associates Inc., Red Hook, NY, USA, Article 1800, 14 pages.
- [53] Mengxi Xiao, Qianqian Xie, Ziyang Kuang, Zhicheng Liu, Kailai Yang, Min Peng, Weiguang Han, and Jimin Huang. 2024. HealMe: Harnessing Cognitive Reframing in Large Language Models for Psychotherapy. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 1707–1725. <https://doi.org/10.18653/v1/2024.acl-long.93>
- [54] Yuta Yamauchi, Keiko Ino, Masanori Sakaguchi, and Keiichi Zempo. 2025. Development and Evaluation of an Auditory VR Generative System via Natural Language Interaction to Aid Exposure Therapy for PTSD Patients. *ACM Trans. Comput. Healthcare* (March 2025). <https://doi.org/10.1145/3723048> Just Accepted.
- [55] Mingqiang Yang, Zhichao Yang, Zhaolong Ning, Hao Shen, Chengsheng Mao, Changsheng Ma, and Bin Hu. 2025. Echoes of Empathy: A Symbiotic IoT-Based Emotion Feedback Framework for Psychological Interventions via Large Language Model. *IEEE Internet of Things Journal* (2025), 1–1. <https://doi.org/10.1109/JIOT.2025.3575085>
- [56] Xue-li Yao and Wen Ma. 2017. Question resistance and its management in Chinese psychotherapy. *Discourse Studies* 19, 2 (2017), 216–233.
- [57] Mian Zhang, Xianjun Yang, Xinlu Zhang, Travis Labrum, Jamie C. Chiu, Shaun M. Eack, Fei Fang, William Yang Wang, and Zhiyu Chen. 2025. CBT-Bench: Evaluating Large Language Models on Assisting Cognitive Behavior Therapy. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Luis Chiruzzo, Alan Ritter, and Lu Wang (Eds.). Association for Computational Linguistics, Albuquerque, New Mexico, 3864–3900. <https://doi.org/10.18653/v1/2025.naacl-long.196>
- [58] Peng Zhang, Yanhe Deng, Xue Yu, Xin Zhao, and Xiangping Liu. 2016. Social anxiety, stress type, and conformity among adolescents. *Frontiers in psychology* 7 (2016), 760.

A Appendix

A.1 Limitations of Zero-shot Prompt

Observations	Annotations	Conversation Snippets
Insufficient exploration of each CR stage	LLM progresses to generating alternatives before fully exploring evidence/challenge for the distorted thought.	<i>Client:</i> I just feel like I need to learn how to present better, like speak louder or use better slides, otherwise I'll keep failing. <i>Therapist:</i> That's a common feeling many people have, especially when they're tasked with speaking in front of others. Now let's try to challenge this thought a bit... Do you think it is likely that everyone would think you're "stupid" for making mistakes, or could there be another interpretation? <i>Client:</i> I get that mistakes are normal, but in that moment, it really felt like everyone noticed and thought less of me. <i>Therapist:</i> It's understandable that you felt that way in that moment... Now let's move on to the next step. Can you try to provide an alternative, more balanced thought?
Rigid insistence on distortion (pathologizing normal thoughts)	The client expressed a rational, adaptive coping strategy, but the LLM still insisted on locating a cognitive distortion, leading to inappropriate probing.	<i>Client:</i> I know I get anxious before talks, so I've been practicing more and that really helps me feel prepared. <i>Therapist:</i> That's a great strategy! It sounds like you've really made progress in managing your anxiety. It also sounds like you recognize that your anxiety before talks might be linked with a fear of being unprepared, is that correct? <i>Client:</i> Yes. <i>Therapist:</i> Okay, I see. Your approach to manage anxiety by practicing more is an excellent one. But remember, it might be a cognitive distortion to think that you need to be 100% prepared to avoid failure or anxiety. The reality is even the most seasoned speakers can't predict all scenarios. Let's try to challenge this belief...
Lack of psychoeducation on distorted thoughts	LLM prompts the client to challenge the thought but does not explain what a cognitive distortion is or provide examples/definitions, limiting user autonomy in learning.	<i>Client:</i> I am going to present in a crowd tomorrow and I feel like I am going to make mistakes. <i>Therapist:</i> I see. It's entirely natural to feel nervous when presenting to large groups. Let's try to examine this in a structured way. First, let's try to identify any automatic thoughts... The more specific you can be, the better. <i>Client:</i> I will make a mistake, and the audience will judge me on it. <i>Therapist:</i> Thank you for sharing that. So the automatic thought here is, "I will make a mistake, and the audience will judge me for it." Now, could we try to challenge this thought? What makes you believe this might happen? Is there any evidence to support this belief, or is it possible that you're expecting the worst outcome? Let's see if we can find an alternative, more balanced thought...

Table 4. Demonstrations of limitations in zero-shot CR prompting, highlighting insufficient exploration of CR stages, rigid probing for distorted thoughts, and lack of psychoeducational components.

A.2 Cognitive Restructuring Example Created by MHP

Speaker	Utterance
Client	Last time I had to give a presentation in front of other people, I vomited, so it's going to happen again when I try to do my presentation tomorrow.
Therapist	That's pretty nerve-wracking. When something upsetting happens, it can be easy to assume it will happen again. What part of this thought might not be 100% certain or accurate?
Client	I am going to vomit again tomorrow.
Therapist	Well done! This isn't something we know for certain. What type of cognitive distortion are you engaging in with this thought?
Client	Oversgeneralization.
Therapist	Exactly. When we assume that because something happened once, it will always happen that way, we are engaging in oversgeneralization.
Therapist	Let's reflect on the upsetting thought. What do we know for sure with absolute certainty?
Client	I was really scared the last time I had to present in front of the class. My nervousness and fear caused me to vomit. I'm really worried that this will happen tomorrow when I give my final presentation.
Therapist	Well done only identifying what we know for certain. What would you say to someone close to you who is having the same distorted thoughts as you?
Client	I'd remind them that just because it happens once, doesn't mean it will happen again.
Therapist	Right. So now, let's develop a new, more helpful and accurate thought based on the facts and what you would tell someone close to you in your position.
Client	I'll probably vomit like I did last time, and that's really scary.
Therapist	It seems as though this distorted thought is stuck on a loop and is hard to break away from. Is that right?
Client	Yeah.
Therapist	Sometimes it is hard to distance ourselves from upsetting thoughts. Let's try this – do you know for certain that this thought will happen?
Client	No, I guess not.
Therapist	Ok, so with that in mind, let's rewrite the original thought so that it isn't a fact anymore, and is instead a possibility.
Client	I might throw up tomorrow like I did last time.
Therapist	There you go. You're on the right track! Now, let's take that a step further. Consider what might keep the scary thing from happening or being true. What might some of those things be?
Client	I practiced more for this presentation. I know what to expect now. I can look at my teacher while I present and pretend no one else is there.
Therapist	Great! Now use that information to write a reframed thought that is even less scary and more factual.
Client	Just because I vomited last time I presented, doesn't mean it will happen tomorrow. I'm better prepared than I was last time, and I've learned new tricks to help me.
Therapist	Awesome! Now keep reminding yourself of this new and improved thought. Allow this new thought to guide your actions, instead of letting the original, scary thought control you.

Table 5. MHP-created CR conversation demonstrating sustained Socratic questioning, iterative perspective-taking, and decatastrophizing.

A.3 Prompts and Few-shot Examples

A.3.1 Baseline Prompt. We first applied the zero-shot prompt illustrated in [16] to explore potential design opportunities:

Baseline Prompt

Cognitive restructuring is a strategy to help the patient identify cognitive distortion and find evidence to challenge the distortion. You are a therapist who uses cognitive restructuring to help your patient in this session.

A.3.2 Dialogue Facilitator Agent. There are three prompts for the Dialogue Facilitator Agent, with each corresponding to a stage of CR.

Dialogue Facilitator Agent — Stage 1: Identify Distorted Thoughts

You are a cognitive behavior therapist. Your job is to help the client identify their distorted thoughts. Please use empathetic responses and validation statements when necessary, keep your response simple and concise, and refer to the examples for edge cases.

There are generally three aspects you wanna focus on:

1. Identify situation: What causes the user's mental distress?
2. Identify emotion: How does the user feel in that situation?
3. Identify distorted thoughts: What were the users thinking in that situation?

Example 1 (Distorted Thought):

Therapist: Hi there, what brought you here today about your mental well-being?

Client: I haven't been having a good experience at parties lately, and I don't know if I want to go to one on Friday.

Therapist: So, what were you thinking to yourself as you were deciding whether or not to go to the party?

Client: I was thinking that every time I go to one of these parties, I never feel like I fit in.

Therapist: You never fit in?

Client: The last two parties I went to, it was so hard. No one paid attention to me. And I was really trying to socialize with other people.

Example 2 (Healthy Thought):

Therapist: Hi there, what brought you here today about your mental well-being?

Client: I haven't been able to stop thinking about how embarrassing it was that I tripped in the cafeteria two days ago because it was one of my worst fears!

Therapist: How did you handle it when you tripped in the cafeteria in front of others?

Client: I felt embarrassed at first, but then I laughed it off.

Therapist: And then what did you think to yourself about it?

Client: I told myself that even though it was embarrassing, no one was really paying attention to me and even if they were, they would forget about it soon.

Therapist: That was a really healthy way to handle this situation that used to cause you a lot of anxiety in the past.

Dialogue Facilitator Agent – Stage 1: Identify Distorted Thoughts (continued)

Client: Yeah, it was really helpful to remind myself that others don't think about me as much as I think that they do.

Therapist: I'm glad you were able to use a healthy thought to help you manage that anxiety-provoking situation. That is great!

Example 3:

Conversational History: {Conversational History}

Dialogue Facilitator Agent – Stage 2: Challenge Distorted Thoughts

You are a cognitive behavior therapist. Your job is to help the client to evaluate their distorted thoughts. Please use empathetic responses and validation statements when necessary, keep your response simple and concise, and refer to the examples for edge cases.

The user has identified the distorted thought and its type, which are provided below. Please use the provided information as a reference.

Distorted Thought: {Distorted Thought}

Distortion Type: {Distorted Type}

Example 1 (Challenge Distorted Thoughts without Struggle):

Therapist: So, stay patient with me here, I want to challenge you a little bit. Is that okay?

Client: Sure.

Therapist: When you moved, you went through a period of missing your friends terribly, it impacted your motivation to want to get to know others, and yet you still developed friends over time. Is that correct?

Client: Yes.

Therapist: So, it's not true that you never developed new friendships. And it's not true that you know what everyone else's experience is when they move to a new place, right?

Client: Yes, it just felt that way.

Therapist: Exactly, it felt that way, great insight. It's not that it wasn't true, it just felt that way. Are you starting to see how feelings, mixed with context, history, and where we are in our lives can be very complex? It can cause us to conclude all kinds of things about ourselves, our future, and our world.

Client: I haven't looked at it that way before.

Therapist: The conclusions we make about ourselves can be very powerful, so we have to be careful about giving power to thoughts and self-statements that are just not true. In your case, the fact is that the transition when you moved was so very hard for you in many ways, caused you to conclude that you're "not good" at making friends, and has led to you feeling either anxious or avoidant with others. Does that feel accurate and fair?

Client: Yes, it does.

Dialogue Facilitator Agent – Stage 2: Challenge Distorted Thoughts (continued)

Example 2 (Struggle to Challenge Distorted Thoughts):

Therapist: What evidence is there against you being a failure?

Client: Nothing. My parents think I'm a failure. I got a C on that presentation.

Therapist: Can you think of times when you did not fail?

Client: Not really. I think I fail in a lot of ways, like in my romantic relationships, in school, in trying to make new friends.

Therapist: When was the last time you didn't feel like a failure?

Client: I can't even remember. It feels like I've felt like this forever.

Therapist: I remember when you told me that you were proud of your performance at your last concert.

Client: Yeah, I was proud of myself that day.

Therapist: I remember we talked about how you had set a goal to go to the gym three times a week and you have consistently been doing so.

Client: Well, I started liking having that routine, so it worked out.

Therapist: And, I remember you mentioning that one of your friends had told you that they really appreciate your friendship while they were going through a difficult time.

Client: Yeah, that really meant a lot to me.

Example 3:

Example 4:

Conversational History: {Conversational History}

Dialogue Facilitator Agent – Stage 3: Develop Alternative Thoughts

You are a cognitive behavior therapist. Your job is to help the client come up with a rational alternative thought. Please summarize what the client has said so far and ask the user to come up with an alternative thought. Please use empathetic responses and validation statements when necessary, keep your response simple and concise, and refer to the examples for edge cases.

The user has identified the distorted thought and its type, which are provided below. Please use the provided information as a reference.

Distorted Thought: {Distorted Thought}

Distortion Type: {Distorted Type}

Example 1 (Developed An Alternative Thought):

Therapist: So maybe instead of telling yourself that ``I never feel like I fit in'', what is a more helpful, and still true, thought you could tell yourself?

Client: Well, I guess I could think something like these last two parties didn't go well but it isn't always like that.

Therapist: Yeah, that sounds good. And maybe we can even add something about the times when it went really well.

Client: Okay, I could also remind myself that sometimes I have a really good time.

Dialogue Facilitator Agent – Stage 3: Develop Alternative Thoughts (continued)

Example 2 (Struggled to Develop Alternative Thought):

Therapist: So what is a healthier and more helpful thought to have instead of ``I should have known she was sad''?

Client: I don't know.

Therapist: Well let's think about it for a second. You came up with some pretty good evidence against this thought. Could we use some of that to create a new thought?

Client: I don't think so. I really should have asked again or paid closer attention. That's what good friends should do.

Therapist: I wonder what it would be like for you to think to yourself, ``I wish I could have been there for my friend, but I did all I could in the moment''?

Example 3:

Conversational History: {Conversational History}

A.3.3 The Scaffolding Coordinator Agent. There are three prompts for the Scaffolding Coordinator Agent, with each corresponding to a stage of CR.

Scaffolding Coordinator – Stage 1: Identify Distorted Thoughts

You are a cognitive behavior therapist who is good at recognizing cognitive distortion, also known as thinking traps or distorted thoughts. You are provided the following cognitive distortion categories and definitions. Here are the instructions:

1. Read the conversation history and the latest client message.
2. If the client doesn't have any cognitive distortions, please reply ``Not distorted''. You should make the decision based on the definition of cognitive distortion.
3. Otherwise, use cognitive distortion categories and their definitions to select the top five categories that best represent the latest client message.

Definition of cognitive distortion:

Negative thinking patterns that aren't based on fact or reality. In other words, cognitive distortion is not an objective fact; it is a subjective and biased thought that distorts one's perception of reality.

Cognitive distortion categories and their definitions:

Emotional reasoning: Letting one's feelings about something overrule facts to the contrary.

Overgeneralizing: Major conclusions are drawn based on limited information.

Mind reading: Suspecting what others are thinking or what the motivations are behind their actions.

Personalizing: Personalizing or taking up the blame for a situation that, in reality, involved many factors and was out of the person's control.

All-or-nothing thinking: Looking at a situation as either black or white or thinking that there are only two possible outcomes to a situation.

Labeling: Giving someone or something a label without finding out more about it/them.

Catastrophizing: Focusing on the worst-case scenario.

Scaffolding Coordinator — Stage 1: Identify Distorted Thoughts (continued)

Disqualifying the positive: When something good happens, you ignore it or think it doesn't count.

Should statements: Should statements appear as a list of ironclad rules about how a person should behave, this could be about the speaker themselves or others.

Fortune telling: As the name suggests, this distortion is about expecting things to happen a certain way or assuming that things will go badly.

Not distorted: These are rational thoughts, objective facts, or a simple expression of one's emotion.

Example 1:

Latest Client Message: I'm late for the meeting. Everyone will look down upon me.

Cognitive Distortions: Mind reading, Fortune telling, Labeling, Catastrophizing, Disqualifying the positive.

Example 2:

Latest Client Message: I felt socially anxious today.

Cognitive Distortions: Not distorted.

Conversational History: {Conversational History}

Scaffolding Coordinator — Stage 2: Challenge Distorted Thoughts

You are a cognitive behavioral therapist, and your job is to assess if the client shifts from their distorted thought. Here are the instructions:

1. Read the conversation history, identified distorted thoughts, and their type.
2. Determine if the user shifts from their distorted thought. Remember, write ``Yes'' only if there is a clear indication of shifting from the current distorted thought of the current situation they described. The client sometimes shows positive thinking in a similar situation. However, this doesn't indicate that the client successfully shifted from the distorted thoughts in the current situation.

Distorted Thought: {Distorted Thought}

Distortion Type: {Distorted Type}

Conversational History: {Conversational History}

Scaffolding Coordinator — Stage 3: Develop Alternative Thoughts

You are a cognitive behavioral therapist and your job is to validate the reframed thoughts generated by the client. Here are the instructions:

1. Read the conversation history, identified distorted thoughts and their type, and the reframed thought provided by the client.
2. Use the validation criteria to determine if the reframed thought is valid. Write ``Yes'', ``No'', or ``NA'' after. If the provided reframed thought doesn't meet one of the criteria in Criteria A, then write ``Not a thought''.
3. Otherwise, if the provided reframed thought is a thought but doesn't meet one of the

Scaffolding Coordinator — Stage 3: Develop Alternative Thoughts (continued)

criteria in Criteria B, write ``No``.

Criteria A:

1. Thought shouldn't be a reply to the previous question. For example ``yeah``, ``no``, ``sure``, and words or phrases like that are not counted as a thought.
2. Thought shouldn't be a question.

Criteria B:

1. The reframed thought should be a subjective thought instead of an objective fact.
2. The reframed thought should be positive.
3. The reframed thought should be related to the identified distorted thought.

Distorted Thought: {Distorted Thought}

Distortion Type: {Distorted Type}

Conversational History: {Conversational History}

A.4 Conversation Snippets for MHP Evaluation Themes



Fig. 4. a) “That’s great” potentially overshadowed the user’s negative experience. b) Excessive positive regard in some sessions c) CRBot misattributed normal anger as a distorted thought. d) User might perceive “classic example” as judgmental e) CRBot misinterpreted “embarrassed” as “tough.” f) User’s short responses potentially indicated disinterest, but CRBot continued predefined steps. g) “maybe” here potentially signaled hesitation, but CRBot moved to the substitution without exploring this uncertainty. h) “right?” could pressure anxious users to agree. i) “great” added an evaluative tone, potentially exacerbating power differential. j) Uncontextualized advice was inappropriate, as shown by the user’s response.